

LONG BEACH, CA, USA

WMO: 722970

Lat: 33.812N

Lon: 118.146W

Elev: 10

StdP: 101.21

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23129

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	5.4	6.6	-11.1	1.5	20.7	-7.7	2.0	18.9	9.0	13.8	7.9	15.3	0.8	300	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.4	33.2	18.9	31.2	18.9	29.5	18.6	22.3	28.3	21.5	27.3	20.8	26.3	3.7	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.4	15.1	24.4	19.6	14.3	24.0	18.8	13.6	23.5	65.2	28.3	62.5	27.2	60.1	26.5	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.4	6.1	5.4	2.9	38.1	1.1	2.2	2.1	39.7	1.4	40.9	0.8	42.1	-0.1	43.7		
(5)			0.4	23.5	1.3	1.3	-0.5	24.4	-1.3	25.2	-2.0	25.9	-2.9	26.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.4	14.2	14.2	15.8	17.0	18.6	20.4	22.8	23.5	22.8	20.1	16.8	13.8
	DBStd	4.24	2.82	2.69	2.55	2.71	2.24	2.00	2.01	2.23	2.77	2.81	2.86	2.39
	HDD10.0	4	2	1	0	0	0	0	0	0	0	0	0	1
	HDD18.3	633	131	119	87	57	22	3	0	0	0	11	63	142
	CDD10.0	3058	133	119	179	211	265	313	397	419	384	314	203	119
	CDD18.3	644	3	3	8	17	29	66	138	161	134	66	16	1
	CDH23.3	3832	73	63	103	173	166	237	643	897	801	466	178	32
	CDH26.7	1243	14	17	30	59	58	58	165	292	297	188	61	4
Wind	WSAvg	2.3	1.8	2.2	2.4	2.7	2.7	2.6	2.6	2.5	2.3	2.0	1.7	1.8
Precipitation	PrecAvg	316	70	81	45	19	7	2	1	0	3	12	22	53
	PrecMax	632	325	330	152	51	54	22	11	2	40	113	57	249
	PrecMin	121	0	0	0	2	0	0	0	0	0	0	0	0
	PrecStd	153	82	75	44	14	12	5	2	0	8	23	18	53
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.6	29.6	30.4	32.3	32.4	31.9	32.6	34.6	36.1	35.8	32.2	26.7
		MCWB	13.7	14.1	15.1	15.4	15.6	19.5	20.6	21.0	20.9	17.0	15.6	12.3
	2%	DB	25.4	24.9	26.2	27.9	27.5	27.9	30.2	31.7	32.5	31.6	28.1	23.5
		MCWB	12.6	13.2	13.7	14.8	17.1	19.1	20.3	20.6	20.5	16.6	13.8	12.3
	5%	DB	22.6	21.6	23.0	24.7	24.4	25.8	28.5	29.8	30.0	28.2	24.8	21.0
		MCWB	12.3	12.6	13.6	14.5	16.9	18.6	20.0	20.3	19.9	16.4	13.6	11.7
	10%	DB	19.6	19.1	20.5	21.9	22.3	24.1	26.8	28.0	27.7	25.4	22.1	18.8
		MCWB	12.1	12.6	13.7	14.3	16.3	18.0	19.8	20.2	19.6	16.9	14.0	11.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.5	16.6	17.9	18.6	20.3	20.8	22.5	23.3	23.8	21.4	18.4	16.6
		MCDB	20.0	22.3	24.3	25.4	25.7	27.9	28.8	30.5	29.9	26.3	23.3	19.6
	2%	WB	15.4	15.5	16.5	16.9	18.7	20.0	21.6	22.2	22.3	20.1	17.1	15.5
		MCDB	18.5	19.8	21.5	23.1	23.9	26.3	27.3	28.3	28.2	25.8	21.3	18.5
	5%	WB	14.7	14.7	15.5	15.9	17.7	19.1	20.9	21.4	21.3	19.0	16.3	14.6
		MCDB	18.0	18.4	19.8	21.2	22.5	24.6	26.2	27.1	27.2	24.5	20.7	17.5
	10%	WB	13.9	13.9	14.8	15.1	16.8	18.5	20.3	20.7	20.5	18.0	15.5	13.8
		MCDB	17.5	17.4	18.7	19.8	21.2	23.2	25.3	26.1	26.1	22.7	19.7	16.6
Mean Daily Temperature Range	5% DB	MDBR	10.6	9.5	9.2	9.3	7.8	7.7	8.6	9.4	9.5	9.7	10.5	10.4
		MCDBR	15.9	14.7	14.6	14.9	12.2	11.0	11.1	12.2	13.2	15.6	15.7	14.9
		MCWBR	7.0	6.5	5.9	5.5	4.5	4.1	3.9	4.2	4.3	5.6	6.5	7.2
	5% WB	MCDBR	9.9	10.0	10.0	11.2	10.1	10.0	9.8	10.7	11.3	11.2	10.8	9.3
		MCWBR	5.4	5.3	4.9	4.9	4.1	3.9	3.7	4.0	4.1	4.7	5.5	5.6
Clear-Sky Solar Irradiance	taub	0.339	0.352	0.377	0.375	0.376	0.392	0.392	0.376	0.362	0.370	0.321	0.321	
	taud	2.494	2.469	2.369	2.389	2.417	2.372	2.440	2.470	2.501	2.450	2.588	2.587	
	Ebn at Noon	871	897	897	913	910	890	888	900	901	863	889	875	
	Edh at Noon	86	98	116	119	117	122	114	109	101	98	78	75	
All-Sky Solar Radiation	RadAvg	3.17	4.06	5.42	6.62	7.10	7.39	7.39	7.04	5.95	4.53	3.52	2.87	
	RadStd	0.22	0.39	0.30	0.34	0.49	0.46	0.41	0.23	0.26	0.34	0.22	0.22	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	+0.43	N/A	-2.62	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (54 neighbors)	+0.62	N/A	-1.65	+0.70	N/A	N/A	-112	+220	+107	

Nomenclature: See separate page